Effect Of Porosity On Vortex Formation Time Scales

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Abstract: Vortex rings are a vastly studied phenomenon in fluid mechanics due to their large presence in turbulent flow. They are flow structures generated when a body of water is forced through an outlet, causing a separation in the boundary layer at the opening, resulting in a rotating column of air. Gharib et al. proposed that vortex rings achieve a maximum circulation at a universal formation number. Using a piston-cylinder experimental setup, he described this characteristic number to be a stroke-to-diameter ratio of approximately 4. In this project, the effect of porosity in formation time scales of vortex rings is studied experimentally. By allowing smaller streams to flow across its surface, porous bodies display different aerodynamic properties than a conventional solid plate. These jets interact with wake flow affecting the formation of vortex structures. Thus, we will utilize perforated plates with industry-standard porosity ratios. By accelerating these plates through a water tank, we obtain velocity and vortex flow fields using Particle Image Velocimetry. By varying the porosity of such plate, we seek to identify the critical ratio at which flow transitions from wake to trailing jet behavior. Ultimately, porosity could serve as a potential parameter space to model the formation time scale of vortex rings.

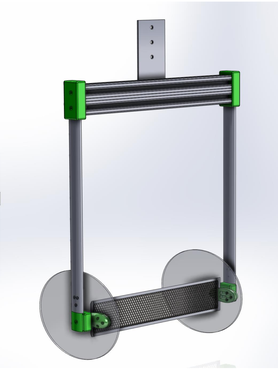
Introduction

Upon the discovery of vortex formation times, researchers have explored the effects of mechanical properties on the universal vortex formation timescale. Kim and Gharib (2010) studied the effects of material flexibility on vortex formation in solid plates of low aspect ratio. They reported strong differences in the vortex rings generated by the translating plate. The flexible plate suppressed some of the upward flow at the tip, resulting in the vortex acquiring a horseshoe shape. Furthermore, the flexible plate presented additional flow in the spanwise direction, resulting in an earlier vortex separation.

Following this line of research, in this project we present the relationship between material porosity and vortex formation. Through experimental analysis we study the development of vortices from perforated plates of high aspect ratio and their interaction with the wake flow. Ultimately, we seek to find how the how the level of perforation of a flat plate affects the vortex formation time scale.

Methods and Materials

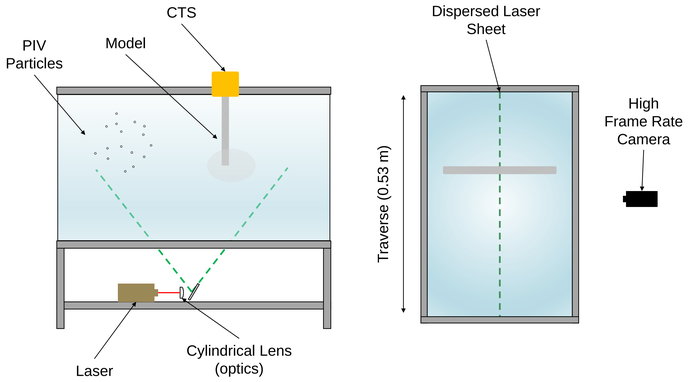
In order to understand the aerodynamic behavior of porous bodies, we perform a series of experiments by translating a perforated plate across a large water tank, allowing us to observe the vortex flow field generated by the moving plate.



Three different perforated plates were studied in this project, selected at industry-standard porosity ratios:

- Solid Plate (0%)
- 30% Open Area
- 60% Open Area

This assembly was designed to interface with the CTS, providing a mounting solution for the perforated plates.



Analysis / Results

Dye Visualization was used to characterize the behavior of the flow across the different porosity cases. This method consists of injecting fluorescent dye into the fluid and illuminating the cross section along the traverse. As the plate begins accelerating, the dye gets mixed in the flow, allowing us to identify the general behavior of the vortex.

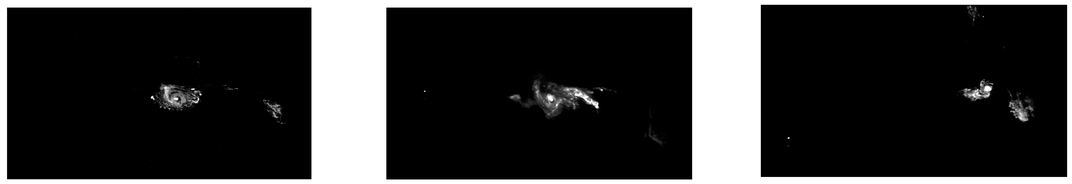


FIG. 1: Dye Visualization at Arbitrary Frame
Solid Plate (left), 30% Open Area, and 60% (middle) Open Area (right)

Given these initial observations, Particle Imaging Velocimetry was used to quantitively asses the dynamics of the flow generated by the accelerating plate. This method consists of seeding the fluid with microscopic reflective particles. A high-power laser illuminates the cross section along the direction of the traverse. By tracking the displacement of each particle over a given time window, we can calculate the velocity at a point in space. Furthermore, this can be extended to a velocity vector field at every point in the frame. Once the velocity field has been obtained, the vorticity was computed by taking the curl of the vector field. Vorticity represents the tendency of the flow to rotate.

Analogous to Gharib et al. (1998), we seek to identify a dimensionless time at which the vortex achieves maximal circulation, marking the formation number. After this point, we expect the vortex to detach from the wake.

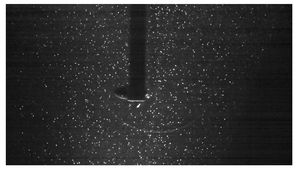


FIG. 2. Raw PIV Capture

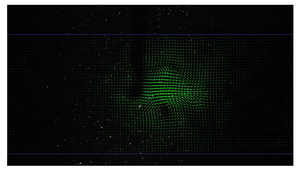


FIG. 3. PIV Lab Vector Field

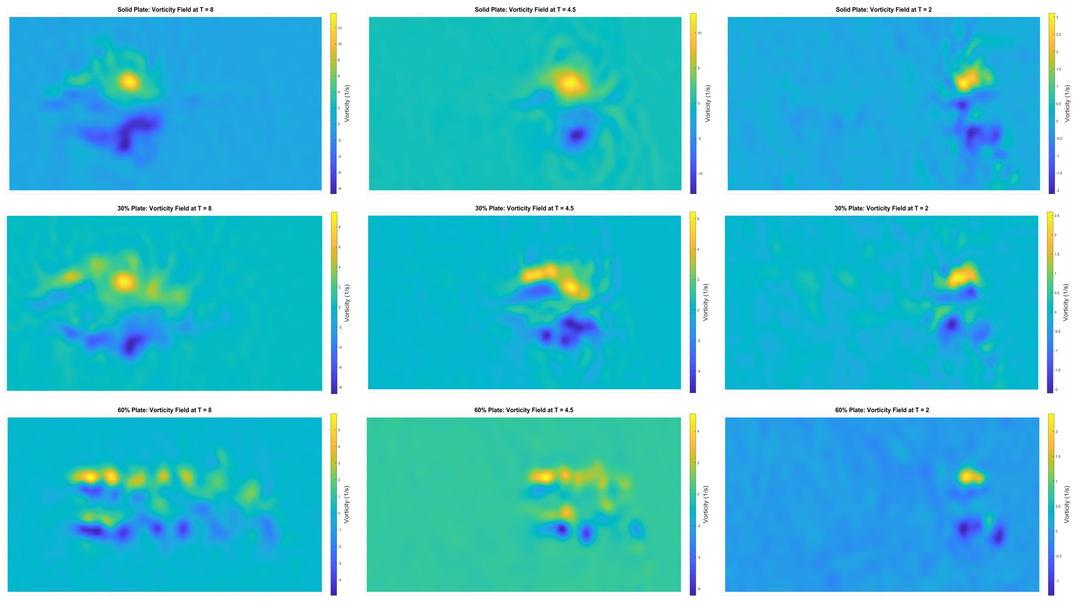


FIG. 4. Vorticity fields for Stroke Lengths T = 2, 4.5, 8
Solid Plate (top), 30% Open Area (middle), and 60% Open Area (bottom)

Conclusion

Given the postprocessed PIV data, we calculated the circulation of the vortex by evaluating the integral $\Gamma = \oint F \cdot ds$, where ds is a closed curve of interest. However, it is crucial that this line integral only encloses the vortex region, even after the vortex is shed. This will ensure that circulation is not computed over unwanted vorticity extraneous to the vortex of interest.

Ultimately, we graph the circulation of each plate across time. We observe that the solid plate achieved a formation number at $T = 6.8$. However, the vortex generated by the perforated plates achieved a formation number at a much earlier time. On the 30% Open Area, we observe a peak in circulation at $T = 3.3$. On the 60% Open Area, we observe a peak in circulation at $T = 2.3$. This behavior may be attributed to the addition of jets flowing across the surface of the perforated plates and their interaction with the trailing wake.

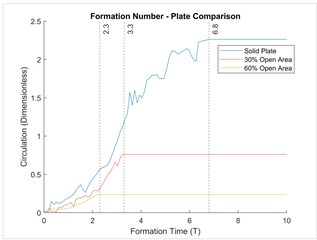


FIG 5. Drag Force vs Non-Dimensional Time

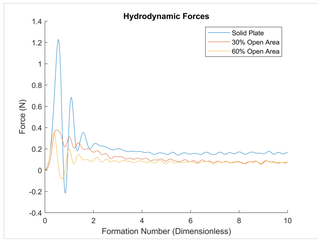


FIG 6. Drag Force vs Non-Dimensional Time

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